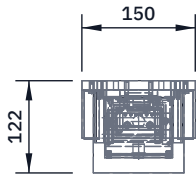


CONCEPT, NOT FOR FABRICATION

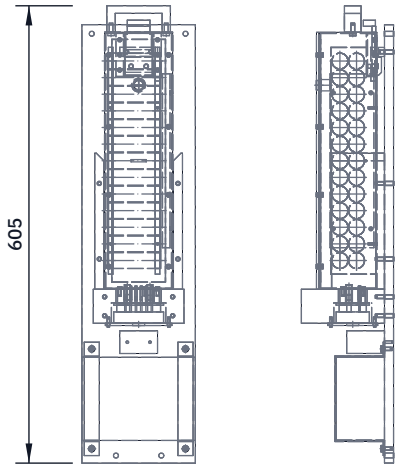
SwapCell · [github.com/BoujeeEnjinia1701/swapcell](https://github.com/BoujeeEnjinia1701/swapcell)

| REV | DESCRIPTION                                                 | DATE       | BY |
|-----|-------------------------------------------------------------|------------|----|
| P1  | Preliminary GA, interface v0.3 (SWC-CAL-001)                | 2026-09-25 | AC |
| P2  | Notes: connector family, R3 derating (SWC-DDR-002)          | 2026-09-25 | AC |
| P3  | Constructable design: latch, fixings, dock (SWC-DDR-003)    | 2026-10-01 | AC |
| P4  | Interface v0.4: handle zone 84 x 43, catch geometry details | 2026-10-02 | AC |



TOP VIEW

Scale 1:10; looking down (along -Z)

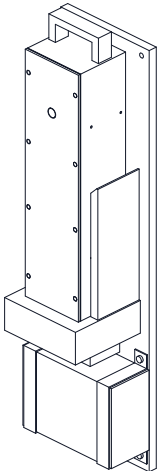


FRONT VIEW

Scale 1:10; looking at the front (along +Y)

RIGHT VIEW

Scale 1:10; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

INTERFACE V0.4 KEY DIMENSIONS (MM)

Pack body 340 x 90 x 80, tolerance +0 / -1.5

Overall length 393 (handle 35, plug 18)

Datum A: connector face; insertion along Z, connector first

Plug centred in width, 8 toward back face from depth centre

Guide faces 340 x 80, 90 apart; receiver clearance 1 per side

Latch pawl 36 wide, 45 below top end, back face

Catch: face 1 above pawl, reach 8, engagement 4 (detail B)

Handle zone 84 x 43 to back face, 35 high (detail A)

Latch class V1: proof 1.72 kN along insertion axis

Vehicle receiver preload 330 N via over-centre lever

Contacts: P1 PACK+, P2 PACK-, S1 SGND, S2/S3 CAN,

S4 WAKE, S5 AUX, S6 INTERLOCK (10 kOhm coded)

Connector: custom keyed shroud, commercial contacts

R3: 20 A open-air mount; derate 50 to 60 C cell temp

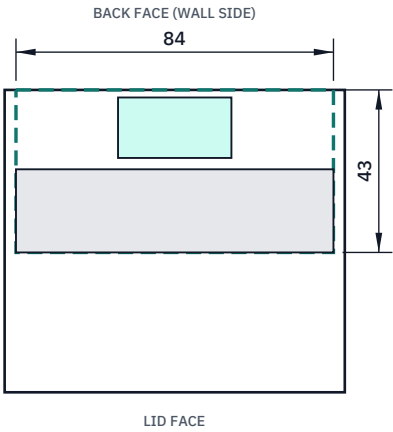
Flush wake button on lid face

Pack mass about 3.0 kg; 13S2P, 46.8 V, 468 Wh

PRELIMINARY, NOT FOR FABRICATION

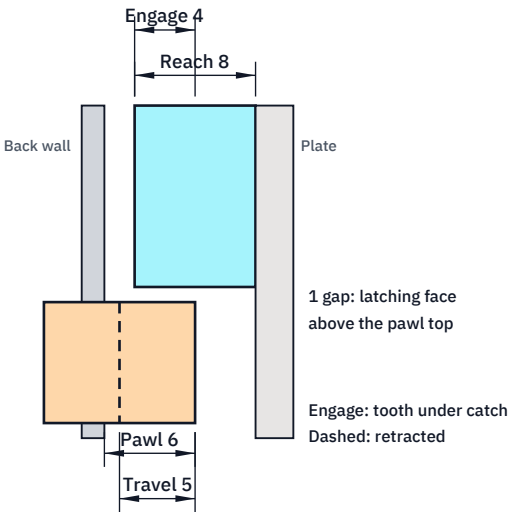
Handle zone (dashed):  
receivers keep it clear

Grey: handle  
Teal: thumb button



DETAIL A: HANDLE ZONE

Scale 1:2; top end, looking down (along -Z)



DETAIL B: CATCH GEOMETRY

Scale 2:1; section, looking at the right side (along -X)

SWAPCELL

[github.com/BoujeeEnjinia1701/swapcell](https://github.com/BoujeeEnjinia1701/swapcell)

General arrangement, pack in wall dock

|                                                                                                                                                    |                    |                  |                           |                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------|---------------------------|---------------------------------------------|
| DRAWING NO.<br><b>SWC-DWG-002</b>                                                                                                                  |                    | REV<br><b>P4</b> | SHEET<br><b>1 of 1</b>    | SIZE<br><b>ANSI B</b>                       |
| SCALE<br><b>1:10</b>                                                                                                                               | UNITS<br><b>mm</b> | PROJECTION<br>   | DATE<br><b>2026-10-02</b> | DRAWN<br><b>Amish Chadha</b>                |
| MATERIAL / NOTES<br><b>Pack: 1.5 mm 5052 Al tray, 3 mm UL 94 V-0 PC sheet lid; dock: 12 mm Al plate, printed shelf and guides. See bom/bom.csv</b> |                    |                  |                           |                                             |
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